Katherine Miner

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I am a Ph.D. student in Computational and Applied Mathematics at the University of Chicago. My work brings mathematical and computational methods to problems in genetics, political science, and law.

Education

University of Chicago 2024 – Present

- Ph.D. in Computational and Applied Mathematics

Massachusetts Institute of Technology (GPA: 5.0/5.0) 2020 – 2024

- B.S. Mathematics
- Minors in Computer Science and Political Science

Research Experience

UChicago Graduate Research with Matthew Stephens May 2025 – Present

- Working in statistical genetics, developing new priors for Empirical Bayes Normal Means and Empirical Bayes Matrix Factorization models to better understand gene expression

UChicago Graduate Research with Steven Kochevar Sep. 2025 – Present

- Developed a mixed-integer optimization model to determine optimal prison placements for accessibility, solved at the state level across the U.S.

MIT Mathematics UROP with Jörn Dunkel Sep. 2023 – May 2024

- Examined how to compress spatial data regarding how DNA is structured within the nucleus of a cell.
- Explored how to create an optimal wavelet to compress 2D and 3D space-filling curves.

MIT Mathematics UROP with Jörn Dunkel Sep. 2022 – Sep. 2023

- Used Virtual Metabolic Human to create a comprehensive network of published biological chemical reactions.
- Built sub-networks based on different metrics such as connectivity, and tested these networks with biological sanity checks.

Imperial College REU with Nick Jones Summer 2023

- Built on existing literature and researched various 3 phase probabilistic models for mitochondrial DNA and their mutation patterns.
- Studied changes in mutation behavior given different starting conditions or intervention methods.

Directed Reading Program Mentee Jan. 2023

- Read "Hopf Monoids and Generalized Permutahedra" by Marcelo Aguiar and Federico Ardila
- Presentation on Hopf monoids of graphs, matroids, and posets, and the calculation of the antipode under mentor Colin Defant

Directed Reading Program Mentee Jan. 2022

- Read chapters from "Statistical Mechanics of Lattice Systems" by S. Friedli and Y. Velenik with mentor Roger Van Peski
- Presentation on the representation of magnetic systems via the Ising and Curie-Weiss models

Mentoring and Teaching

UChicago Math Department Instructor — Fall 2026
- Lecturing one section of introductory calculus (MATH 131) for first-year students

UChicago Math Department Teaching Fellow — Sep. 2025 – June 2026
- Graded homework, gave guest lectures, designed exam problems and led review sessions
- Mentored by three different math professors across the 2025-2026 academic year.
- In Fall '25, Winter '26 and Spring '26, I received a median rating of 5.0/5.0.

MIT Single and Multivariable Calculus Teaching Assistant — Jan. 2023 – May 2024
- Designed recitations, problem sets and practice exams for students, as well as led office hours and recitations.
- In Spring 2024, I received an overall rating of 6.7/7.0.

MIT Single and Multivariable Calculus Undergraduate Assistant — Fall 2022
- Held weekly office hours and review sessions before exams, as well as monitored the online question forums for a class of over 300 students.

MIT Freshman Associate Advisor — Sep. 2021 – May 2024
- Advised a group of 5+ incoming MIT freshmen on classes, extracurriculars, and maintaining a healthy workload.
- Attended monthly training sessions and communicated with other advisors when conflict arose.

Activities and Programs

UChicago CAM Student Committee Representative — Sep. 2025 – Present
- Advocating for department PhD students to department chair and administration
- Elected by fellow students

UChicago CAM Pre-Orientation Student Organizer — Sep. 2025 & 2026
- Coordinating Q&A sessions for incoming Ph.D. students on undergraduate coursework and transitioning to graduate-level mathematics

UChicago CAM & Stats Student Seminar Co-chair — Aug. 2025 – May 2026
- Organized and oversaw weekly student-led seminars to provide opportunities for academic presentation practice

MIT Council for Math Majors (CoMM) Co-chair and Discussion Forums Lead — Aug. 2021 – Jan. 2024
- Co-organized the advocacy group for math majors at MIT
- Worked to build the Course 18 Underground Guide, as well as wrote a proposal for starting student social events in the Undergraduate Lounge
- Worked to create roadmaps of recommended courses for first and second year math majors

Selected Awards and Skills

Charles and Holly Housman Award — 2024
- For skill and dedication in undergraduate teaching

Associate Advisor of the Year — 2023

PEO and Lockport College Women's Club Scholarship Recipient — 2020

Programming Languages: Python, MATLAB, Julia, R, JavaScript